



THAPAR INSTITUTE

Introduction to Cloud Computing

Course Objectives

- Comprehend the basic concepts and architecture of Cloud computing.
- Implement Cloud Services through AWS offerings and Restful web services.
- Apply the knowledge of virtualization through different virtualization technologies.
- Perform operations on data sets using Map Reduce framework, SQL and NO SQL databases.



Syllabus

1. Introduction and Evolution of Computing Paradigms

Architecture, Business Drivers, Main players in the Field, Cluster Computing, Grid Computing, Issues & Challenges

2. Classification of Cloud Implementations

The Elastic Compute Cloud (EC2), Simple Storage Service (S3), Simple Queuing Services (SQS), Bundling Amazon instances, AWS Identity Management and Security

3. Virtualization

Advantages and disadvantages, Types, Migration of processes, Classic & Virtualized Data DCs, VMware vCloud – IaaS, Network virtualization through SDN

4. Cloud based Data Storage

Hadoop Ecosystem (Pig, Hive, Cassandra and Spark), No-SQL, Map-Reduce, Data Replication

5. Related Technologies

Fog & Edge Computing, Cloud for IoT and Big data analytics, Google AppEngine - PaaS, Windows Azure

Lecture Plan

Week 1- 2	Introduction to cloud, Cloud Types and Models, Existing Platforms, Evolution, Benefits and Challenges of Cloud Computing
Week 3- 4	AWS, EC2, S3, SQS, Bundling Amazon instances, AWS IAM and Security, RESTFul Web Services
Week 5-7	Virtualization, Pros & Cons, Types, Migration of processes, Classic & Virtualized DCs, VMware vCloud – IaaS, Network virtualization through Software Defined Networks
Week 8-10	Hadoop Ecosystem (Pig, Hive, Cassandra and Spark), No-SQL databases, Map- Reduce framework, Data Replication, Shared access to data stores.
Week 11-12	Fog Computing and Edge Computing, Usage of Cloud for IoT and Big data analytics, Overview of Google AppEngine - PaaS, Windows Azure

List of Experiments*

Sr. No	Description
1	Learn Python environment and programming in Colab.
2	Programming in Colab environment
3	Realizing AWS Academy and programming environment.
4	Elastic Compute Cloud (EC2)
5	Amazon Simple Storage Service (Amazon S3)
6	Simple Queuing Services (SQS)
7	Using PySpark on Dataproc Hadoop Cluster in Google cloud platform
8	Deploy an application using Elastic Beanstalk
9	Setting up a cloud environment on local server and deploying a virtual machine
10	Implementing storage virtualization and testing different storage virtualization techniques

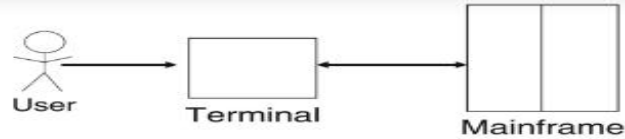
Marks Distribution

COMPONENT	DURATION	MARKS
Quiz	Before MST	15
Mid Semester Examination	September	30
Lab Assessment (Project)	Before EST	15
End Semester Examination	December	40

Computing Paradigms

Phases

1. Mainframe Computing



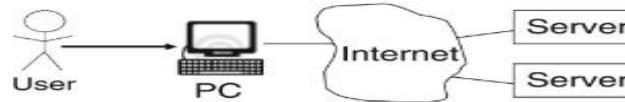
2. PC Computing



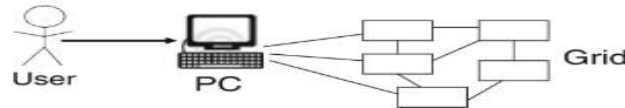
3. Network Computing



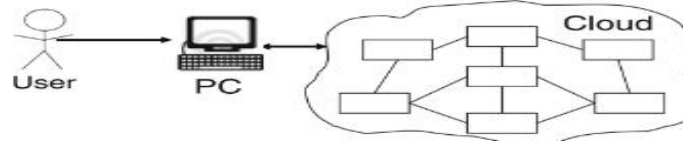
4. Internet Computing



5. Grid Computing

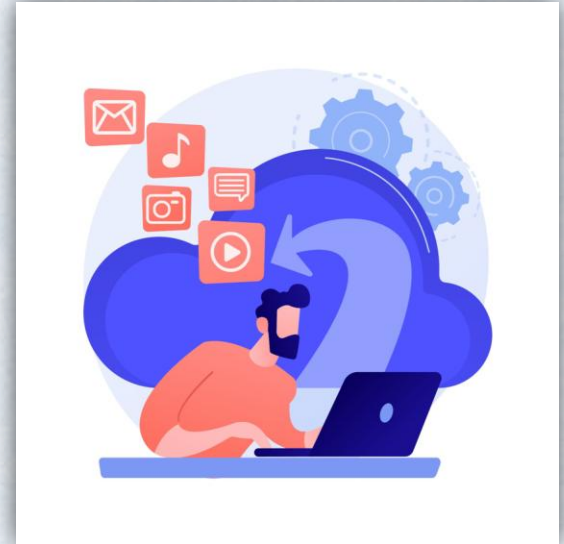


6. Cloud Computing



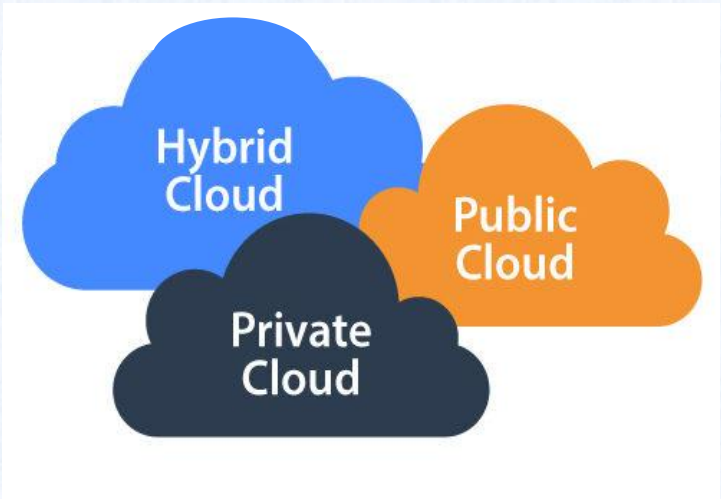
Cloud Computing Overview

- **Definition of cloud computing:** Delivery of on-demand computing resources over the internet.
- **Key features:** On-demand self-service, broad network access, resource pooling, rapid elasticity, measured service.
- **Benefits:** Cost savings, scalability, flexibility, accessibility, reliability, and security.



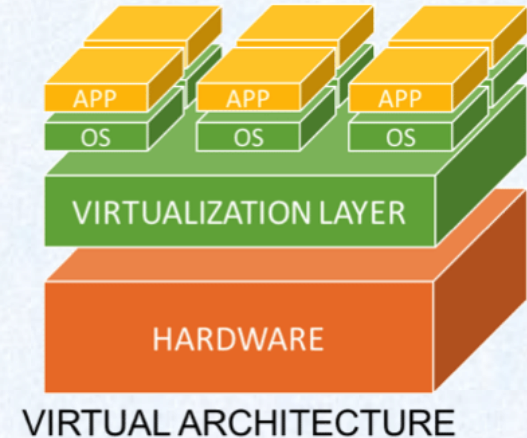
Cloud Deployment Models

- Public Cloud, Private Cloud, Hybrid Cloud, Community Cloud.
- Differences between them.
- Factors to consider when choosing a deployment model.
- Examples of organizations that use each model.



Virtualization Overview

- **Definition:** the process of creating a virtual version of something, such as a server or storage device
- **Advantages:** improved hardware utilization, flexibility, scalability, and cost savings
- **Types:** Server virtualization, network-virtualization, application virtualization, storage virtualization, and desktop-virtualization



THANK YOU